FOR A FLAT RING MAP AND A FINITE IDEAL THE GLUING CATEGORY IS ABELIAN AND THE CANONICAL FUNCTOR IS EXACT AND COMMUTES WITH COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'Details omitted.' inside the proof; that sentence is kept verbatim and marks the one step left to the reader.

Allowed standard prerequisites (editorial).

- The gluing category Glue(R -> S, f_1, ..., f_t) and the canonical functor Can. the objects and functor named in the statement at source lines 24746-24747. Fixes the category whose abelianness and whose functor's exactness are being asserted.
- Flatness of R -> S and of the localizations R_{f_i}, and commutation of kernels, cokernels and colimits with tensor products. used at proof lines 24761-24764. Is the single input that makes the componentwise construction of kernels and cokernels valid and that gives exactness and colimit commutation.

Lemma 0.2. *Assume $\varphi : R \to S$ is a flat ring map and let $I = (f_1, \ldots, f_t) \subset R$ be an ideal. Then Glue$(R \to S, f_1, \ldots, f_t)$ is an abelian category, and the functor Can is exact and commutes with arbitrary colimits.*

Proof. Given a morphism $(\varphi', \varphi_i) : (M', M_i, \alpha_i, \alpha_{ij}) \to (N', N_i, \beta_i, \beta_{ij})$ of the category Glue$(R \to S, f_1, \ldots, f_t)$ we see that its kernel exists and is equal to the object $(\mathrm{Ker}(\varphi'), \mathrm{Ker}(\varphi_i), \alpha_i, \alpha_{ij})$ and its cokernel exists and is equal to the object $(\mathrm{Coker}(\varphi'), \mathrm{Coker}(\varphi_i), \beta_i, \beta_{ij})$. This works because $R \to S$ is flat, hence taking kernels/cokernels commutes with $- \otimes_R S$. Details omitted. The exactness follows from the R-flatness of R_{f_i} and S, while commuting with colimits follows as tensor products commute with colimits. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.